

Healthcare AI QA Engineering Platform

Step 0.3 - Engineering Standards & Architecture Contract

Objective

Define project-wide engineering standards and architecture contracts before feature implementation.

Architecture

Defined modular boundaries: frontend, automation, rag, mcp, agent, mock-systems, knowledge, docs.

Technology Stack (Planned)

- Frontend: React, TypeScript, Vite, React Router
- Automation: Playwright, Cucumber BDD, Gherkin, POM
- AI: Python, ChromaDB/local vector store, FastMCP, LangGraph

Architecture Decisions

- ADR-001 Frontend-only architecture
- ADR-002 Playwright + Cucumber BDD
- ADR-003 Python for AI/RAG/MCP/Agent
- ADR-004 FastMCP for MCP
- ADR-005 LangGraph orchestration
- ADR-006 Local/vector-store-first RAG
- ADR-007 Synthetic healthcare data only
- ADR-008 Lightweight in-project Jira/Confluence/Jenkins

Engineering Standards

- Coding, architecture, security, testing, Git, and documentation standards consolidated

Security

- No secrets, credentials, tokens, PHI, or policy bypass

QA Automation Standards

- Planned POM, reusable fixtures/hooks, stable locators, deterministic data

AI Architecture

- Planned RAG + MCP + LangGraph with explicit boundaries and observability expectations

Git Standards

- Branch strategy and commit naming conventions documented

Documentation Standards

- Mandatory per-step artifacts including verbatim prompt and response

Interview Talking Points

- Why standards-first execution
- Why frontend-only and synthetic healthcare data policies
- Why FastMCP and LangGraph

Next Steps

- Await Step 0.4 (no frontend/BRD/PRD implementation in this step)